

Python Booleans

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Python Tutorial

[Python HOME](#)[Python Intro](#)[Python Get Started](#)[Python Syntax](#)[Python Comments](#)[Python Variables ▼](#)[Python Data Types](#)[Python Numbers](#)[Python Casting](#)[Python Strings ▼](#)[Python Booleans](#)[Python Operators](#)[Python Lists ▼](#)[Python Tuples ▼](#)[Python Sets ▼](#)[Python Dictionaries ▼](#)[Python If...Else](#)[Python Match](#)[Python While Loops](#)[Python For Loops](#)[Python Functions](#)[Python Decorators](#)[Python Range](#)[Python Lambda](#)[Python Arrays](#)[Python OOP](#)

Booleans represent one of two values: True or False.

Boolean Values

In programming you often need to know if an expression is True or False.

You can evaluate any expression in Python, and get one of two answers, True or False.

When you compare two values, the expression is evaluated and Python returns the Boolean answer:

Example

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```
print(10 > 9)
print(10 == 9)
print(10 < 9)
```

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When you run a condition in an if statement, Python returns True or False:



Example

Print a message based on whether the condition is True or False:

```
a = 200
b = 33

if b > a:
    print("b is greater than a")
else:
    print("b is not greater than a")
```

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Evaluate Values and Variables

The bool() function allows you to evaluate any value, and give you True or False in return,

Example

Evaluate a string and a number:

```
print(bool("Hello"))
print(bool(15))
```

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Example

Evaluate two variables:

```
x = "Hello"
y = 15
```

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```
print(bool(x))  
print(bool(y))
```

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Most Values are True

Almost any value is evaluated to True if it has some sort of content.

Any string is True , except empty strings.

Any number is True , except `0` .

Any list, tuple, set, and dictionary are True , except empty ones.

Example

The following will return True:

```
bool("abc")  
bool(123)  
bool(["apple", "cherry", "banana"])
```

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Some Values are False

In fact, there are not many values that evaluate to `False`, except empty values, such as `()`, `[]`, `{}`, `""`, the number `0`, and the value `None`. And of course the value `False` evaluates to `False`.

Example

The following will return False:

```
bool(False)
bool(None)
bool(0)
bool("")
bool(())
bool([])
bool({})
```

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One more value, or object in this case, evaluates to `False`, and that is if you have an object that is made from a class with a `__len__` function that returns `0` or `False`:

Example

```
class myclass():
    def __len__(self):
        return 0

myobj = myclass()
print(bool(myobj))
```

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Functions can Return a Boolean

You can create functions that returns a Boolean Value:

Example

Print the answer of a function:

```
def myFunction() :  
    return True  
  
print(myFunction())
```

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You can execute code based on the Boolean answer of a function:

Example

Print "YES!" if the function returns True, otherwise print "NO!":

```
def myFunction() :  
    return True  
  
if myFunction():  
    print("YES!")  
else:  
    print("NO!")
```

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